

Experiment -2

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Experiment: Study of Machine Learning Dataset Repositories Aim

To study different machine learning dataset repositories such as Kaggle and the UCI Machine Learning Repository, and to examine a student-related dataset available in CSV format.

Objective

- To learn about popular sources of machine learning datasets.
- To compare Kaggle and the UCI Machine Learning Repository.
- To understand the attributes of a student-related CSV dataset.

Introduction

Machine learning algorithms require datasets for training, validation, and testing. These datasets consist of records representing observations along with their corresponding attributes. Public repositories provide datasets for academic learning, research, and realworld applications.

Two widely used repositories are Kaggle and the UCI Machine Learning Repository.

1.Kaggle

Description

Kaggle is an online platform designed for data science enthusiasts, students, and researchers. It hosts thousands of datasets from domains such as education, healthcare, finance, sports, and business.

Features

- Extensive collection of datasets.
- Supports CSV, Excel, JSON, SQL, images, and text data.
- Interactive notebooks for data analysis.
- Machine learning competitions.
- Active community for discussions and learning.

2..UCI Machine Learning Repository Description

The UCI Machine Learning Repository is a trusted academic repository maintained by the University of California, Irvine. It contains benchmark datasets commonly used for teaching and research in machine learning.

Features

- Well-organized datasets.
- Detailed documentation for each dataset.
- Supports multiple file formats such as CSV and ARFF.
- Widely cited in research publications.

Sample Dataset Dataset Name

Student Depression Dataset

Dataset Description

The Student Depression Dataset contains information about students' academic performance, lifestyle, and mental well-being. It is useful for studying factors that may influence stress levels and predicting depression among students.

Each record represents one student.

Sample CSV Data

Student_ID	Age	Gender	CGPA	Study_Hours	Sleep_Hours	Academic_Pressure	Financial_Stress	Depression
201	19	Female	8.6	5	7	High	Low	Yes
202	20	Male	7.8	4	6	Medium	Medium	No
203	18	Female	9.1	6	8	Low	Low	No
204	21	Male	6.9	7	5	High	High	Yes

Description of Attributes

Attribute	Description	Data Type
Student_ID	Unique student identifier	Integer
Age	Age of the student	Integer
Gender	Gender of the student	Categorical
CGPA	Cumulative Grade Point Average	Decimal
Study_Hours	Daily study hours	Integer
Sleep_Hours	Average sleep per day	Integer
Academic_Pressure	Academic workload level	Categorical

Financial_Stress	Financial stress level	Categorical
Depression	Depression status (Yes/No)	Categorical

Result

The experiment explored two major machine learning dataset repositories: Kaggle and the UCI Machine Learning Repository. A Student Depression Dataset in CSV format was analyzed to understand its structure, attributes, and applications in machine learning. The study demonstrated how publicly available datasets can be used for data analysis, prediction, and research.